

Instructor: Date:

Learner: Regional Branch:

Program: Center:

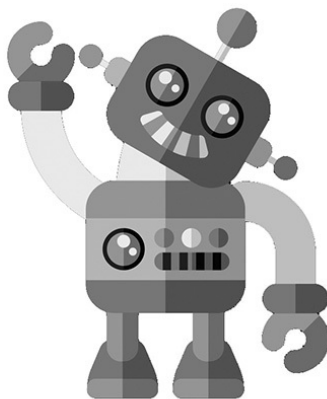
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Let's Start!

The robots are coming!



Learning Outcome

Explain about technology and technical problems in communicative situations.

Ask your partner:

- What technological gadgets do you own?
- If you could have a robot, would you want one? Why or why not?

Can Do Statements

- I can talk about technology and gadgets.
- I can express complex statements using words like *who*, *that* or *which*.
- I can talk about wishes and imaginary situations.





Let's Explore!

I **wish*** ...

* People often "*wish*" on the first star they see at night, or on a coin they throw in a fountain.



Instructions: Read the text aloud.

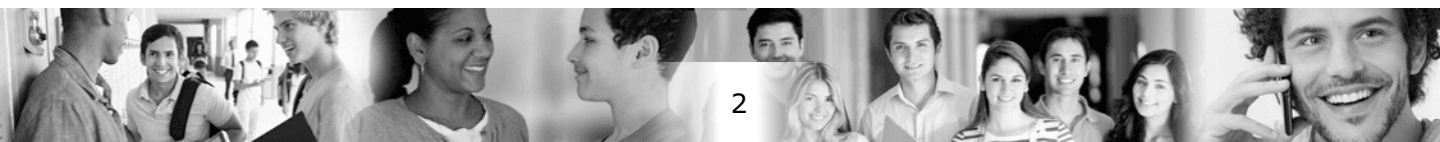
Robotic workers

If you looked up the word robot, you'd find it was first coined by Karel Capek in 1921. The word robot comes from robota, which means serf laborer. Capek's robots were humanoid creations that were used as workers.

If you had a robotic worker at your company today, what would it look like? Would it be a device that resembled a human? Probably, it wouldn't. Today's robots are fashioned to suit their functional purpose. They perform repetitive, dangerous tasks which humans cannot or prefer not to do.

Surgeons operate using robotic arms which hold surgical instruments. In Japan, which is the world leader in robotics, they populate automobile plants, painting, welding, and inspecting. Robotic units assist in bomb disposal. They also help the physically disabled.

So far, they are easily spotted. If you saw one, you would never mistake it for human. However, some scientists still envision human-like robots that have sensors to see and hear, motors that simulate human muscles, and microprocessors that "think". They wish that they could produce robots like that now. Don't you?





Let's Practice!

How would you describe robots?

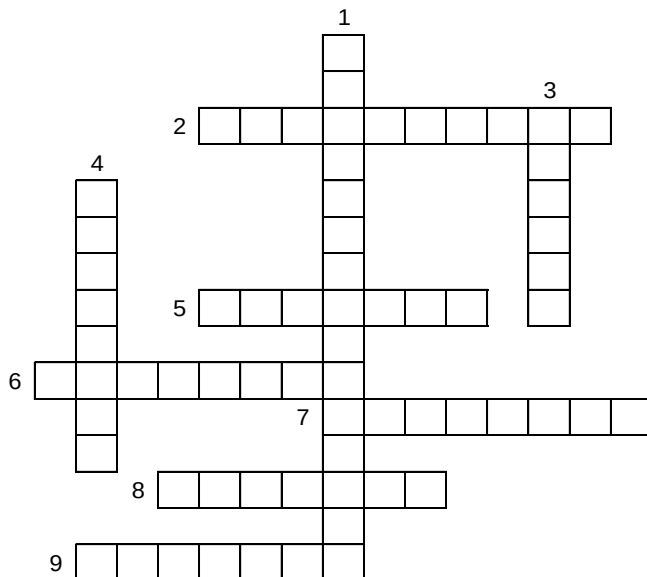
Instructions: Complete the crossword. First, read the clue. Then, write your answer.

Down

1. They will have m_____ that 'think'.
3. Robotic units will a_____ in bomb disposal.
4. Today robots are generally not h_____.

Across

2. Today robots are generally not h_____.
5. Surgeons use r_____ arms.
6. They p_____ automobile plants, painting, welding, and inspecting.
7. In the future robots will have motors that s_____ muscles.
8. Now robots p_____ repetitive, dangerous tasks.
9. They will have s_____ to see and hear.



Let's Practice!

What do robots do?

Instructions: Draw a line to match the sentences in Column A and Column B. When you complete the sentences, read the sentences aloud with your partner. Check your pronunciation with your instructor.

COLUMN A

COLUMN B

1. The word robot comes from robot,
2. Capek's robots were humanoid creations
3. They perform repetitive, dangerous tasks
4. Surgeons operate using robotic arms
5. They are usual in Japan,
6. However, some scientists still envision human-like robots

- a. which is the world leader in robotics.
- b. which means serf laborer.
- c. that were used as workers.
- d. which humans cannot or prefer not to do.
- e. that have sensors to see and hear.
- f. which hold surgical instruments.





Extension Activity

Explain what a robot would look like and what it could do if you built it.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Write a 100-word description of your personal robot.
- Describe what it would look like.
- Say what functions you would like it to perform.
- Explain why you think these things are important.
- Read your description aloud to yourself or to a friend.

Self-Assessment

Yes

No

I write about a robot using some technological vocabulary.

☐ ☐

I use complex statements with words like *who*, *that*, *which*.

☐ ☐

I write about imaginary situations with my personal robot.

☐ ☐

I share my description.

☐ ☐


Target Words

- **to coin** (verb): to invent a new expression
e.g. The word robot was coined by Karel Capek.
- **serf** (noun): someone who worked on the land for the landowner in medieval times
e.g. The word robot means serf laborer.
- **to envision** (verb): to imagine for the future
e.g. Scientists still envision human-like robots.
- **humanoid** (adjective): behaving and looking like a human
e.g. Capek's robots were humanoid creations.
- **device** (noun): a machine invented for a specific purpose
e.g. When you think of a robot, is it a device that resembles a human?
- **to fashion** (verb): to make; to build
e.g. Today's robots are fashioned to suit their functional purpose.
- **robotic** (adjective): relating to a robot
e.g. Surgeons operate using robotic arms which hold surgical instruments.
- **to populate** (verb): to occupy; inhabit
e.g. Robots populate automobile plants.
- **to weld** (verb): to join pieces of metal together with extreme heat
e.g. Robots weld auto parts together.
- **sensor** (noun): a device that notes changes in something
e.g. Future robots will have sensors that see and hear.





Learning Points

Subject and Object Relative Clauses

When you want to clarify which person or thing you are talking about, you use a relative clause.

You can begin a relative clause with *who* for people, *which* for things or *that* for either people or things.

You put the relative clause immediately after the noun it talks about.

A subject relative clause describes the subject of the clause.

Example:

*The woman **who/that** lives next door to me brought me a cake.*

The clause - who/that lives next door - refers to the woman. Who/that is the relative pronoun. It is also the subject of the clause. Who lives next door? The woman.

An object relative clause describes the object of the clause.

Example:

*The apartment **which/that** we rented was near the center of the city.*

The clause - which/that we rented - refers to the apartment. Which/that is the relative pronoun. *We* is the subject of the clause. *Apartment* is the object of the clause. What did we rent? The apartment.

Subject Relative Clause	Object Relative Clause
<i>The word robot comes from robota, which means serf laborer. (What does serf mean? Robot.)</i>	<i>The robots that he created became serf laborers. (What did he create? Robots.)</i>
<i>Robots which perform dangerous repetitive tasks are used in manufacturing. (Who performs dangerous tasks? Robots.)</i>	<i>Robots perform repetitive, dangerous tasks which humans cannot or prefer not to do. (What can't humans perform? Dangerous tasks.)</i>
<i>Surgeons operate using robotic arms which hold surgical instruments. (What holds surgical instruments? Robotic arms.)</i>	<i>The robotic arms that surgeons use are highly valued. (What do surgeons use? Robotic arms.)</i>





Learning Points

Talking about Wishes and Imaginary Situations

You use *wish* when you are sorry that things are not different. You hoped for a different past or present.

You use the second conditional to talk about imaginary situations in the present or future.

Wish form:

Wish + that-clause (you can leave "that" out)

They wish that they could produce robots like that now.

or

They wish they could produce robots like that now.

Second conditional form:

If + Past Simple, modal (could, would, should, might)

If you had a robot worker, what would it look like?

Examples:

- *I wish that I had my own robot!*
- *I wish I could buy a robot that looked human.*
- *If you saw a robot today, you would never mistake it for a human.*
- *If you built a robot, what would it look like?*

